Referee report on `mjl_part4_draft5.tex`, with suggestions for the wild integrality problem

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0. Scope

Read in full. Draft 5 (1589 lines; 22 pages; it compiles cleanly).

Compiled numbering of the new results.

- Example 6.3 (data);
- Lemma 6.4 (toral Fourier test);
- Proposition 6.5 (all quadratic tori);
- §6.4 (Steps T and G);
- Proposition 8.3 (rank-two spherical unit);
- Remark 8.4;
- Proposition 8.5 (conditional comparison with Λ);
- Remark 8.7 (identity germs);
- Example 8.8 (numerical ledger).

Versions and sources.

- Draft 5 cites Parts I and II as Draft 8 and Part III as Draft 9. I have Draft 6 of each, and of [JC] I have revised12 but did not read it.
- I could not verify the following against a source:

  JC , Proposition 2.4 (Stone–von Neumann);
  - the exact pointwise form of [FK, §12, Corollary];
  - Li's real-place hypothesis.
- The literature I suggest in §3 is cited from memory and should be checked.

Summary verdict

- *§§2–5, 7.* Unchanged from Draft 3 apart from the points I raised, all of which are handled. One exception: the $u(x)$ paragraph now says "$u(x) = 1$ by definition of the relevant convention". That is acceptable if [FK, §4] defines it so, but it is no longer an argument. Keep the one-line argument from Draft 3 as well.
- *The wild rank-two example (§§6.3–6.4).* This is a genuine and correct new computation. I rechecked every value and every decomposition (§1.1). There is one misstatement in the description of B_d.
- *Proposition 8.5.* It contains a new hypothesis, that nongood ordinary classes have no contribution. This hypothesis is unnecessary, and it rests on a mismatched pairing of classes (§1.2). It should be deleted. As written it also makes the proposition hard to apply with unit elements at almost all places.
- *Proposition 8.3, Remark 8.7, Example 8.8.* Correct. One citation is unverified (§1.3).

The main recommendation. Add the positivity lemma of §2. It is elementary, and it strengthens Proposition 6.1 considerably: *for the unitary matrix $(a_{\rho\tau})$, integrality is equivalent to nonnegativity*, i.e. to every $L_D\Theta_\rho$ being a positive-definite function. This changes what has to be proved, and it makes several of the routes in §3 much shorter.

1. Correctness of the new material

1.1 The rank-two wild example: correct, one slip

I verified the following.

- *The setup.* $F = \mathbb{Q}_3(\zeta_3)$, $|3|_F = 1/9$. $D = E_0 \oplus E_0 j$ with $j^2 = \varpi$ is division, since ϖ has odd valuation and is not a norm from the unramified $E_0 = F(i)$. E_0 is unramified because -1 is not a square in $\mathbb{F}_3$.
- *The unramified torus,* $R_{E_0} = |a^2 + b^2|/|3a^2 - b^2|$. Both values are right. $a^2 + b^2$ is a unit because -1 is a nonsquare. For $b = \varpi u$, $3a^2 - b^2 = 3(a^2 + \zeta_3 u^2)$ with $a^2 + u^2 \not\equiv 0$. So $R \in \{9, 1\}$ as stated, and $R_{E_0} = 1 + 2\,\mathrm{reg}_{C_4} = 3 + 2\chi_1 + 2\chi_2 + 2\chi_3$.
- *The ramified tori.* $R_{E_d} = |a^2 - db^2|/|3a^2 + db^2|$. The three valuation cases give $9, 3, 1$. The decomposition $1 + \mathrm{reg}_{C_2} + \mathrm{reg}_{C_6}$ has the values 9, 3 and 1 on the identity, the even nonidentity elements and the odd elements. The Fourier coefficients $(3, 2, 1, 1, 1, 1)$ are right.
- *Lemma 6.4.* Correct.
  - $\zeta_{\alpha\chi}$ is well defined, using injectivity of P_n on A_E and $\theta(b) = \tilde{\theta}(b^*)$.
  - It is multiplicative, using multiplicativity of the star on the torus.
  - Stone–von Neumann characters are $b_D\zeta$ on C_E and 0 off it.
  - The restriction identity follows.

  Repair: Lemma 6.4 has no status label; it should be **[P]**.
- *Proposition 6.5.* Correct, including the choice of $\tilde{\theta}$ with $\tilde{\theta}(a^*) = 1$ (the star image meets μ_n trivially) and the Galois invariance.

The slip. In the description of $B_d = E_d^\times / F^\times U_1$, the text says that even classes "modulo $F^\times$ are represented by $1 + cj_d$ with $c \in \varpi\mathcal{O}_F$". Those elements lie in U_1, so they are trivial in B_d. It should say $c \in \mathcal{O}_F$, read modulo ϖ. The multiplication law and the additivity of $\bar{c}$ are then right. It would also help to note that $U_1 = \mathcal{O}_F^\times(1 + \mathfrak{p}_{E_d}^2)$.

Consistency check. The torus results are exactly positive-definiteness of the restriction of $L_D\mathbf{1}$ to each covered torus. This is consistent with §2 below, and it is a necessary condition.

1.2 Proposition 8.5: delete the nongood-class hypothesis

The proposition assumes that every *nongood* rational noncentral ordinary class contributes zero, "removing the ordinary classes without a covered counterpart". But in the comparison the ordinary class $[s]$ is paired with the covered class of $s^* = \iota(s)^n$, not with $\iota(s)$.

- This is what hypotheses (i)–(ii) say: the local identities compare $J(s_v^*, \tilde{f}_v)$ with $J(s_v, f_v)$.
- It is also what the proof itself uses, via $\iota(s^n) = \iota(s)^n$.
- The star of any element is good, since $[\tilde{t}^n, \tilde{u}] = [\tilde{t}, \tilde{u}]^n = 1$.
- Theorem 5.2 gives a bijection from *all* noncentral $[s] \in E^\times/K^\times$ onto the good classes.

So every ordinary class has a counterpart, and no extra support condition is needed.

As written, the hypothesis would require the ordinary test function to kill every $[s]$ with $s \notin K^\times E^{\times n}$. That is incompatible with unit elements at almost all places, unless the function is adjusted at some place to vanish on specific classes. *Delete the hypothesis* and the corresponding sentence of the proof and ledger. "Nongood ordinary class" is meaningful only for the lift $\iota(\gamma)$, which is not the partner of γ.

1.3 Other new material

Proposition 8.3. The elementary part is correct:

- $J(\mathrm{diag}(a, b), 1_{K_0}) = 1$ for a, b units;
- both sides vanish when the valuations differ;
- scalar translates contribute the central character.

The final sentence relies on [FK, §12, Corollary] giving a *pointwise* identity $J(x^*, \tilde{f}^0) = J(x, f^0)$. I could not verify that form.

For $r \geq 3$, Lemma 8.2 shows that such a pointwise identity would force $J(\zeta x, 1_{K_0}) = J(x, 1_{K_0})$ on mixed tori. That is a checkable consequence. Either it confirms the pointwise form for $r = 3$, or it shows that FK's identity involves a root sum. Remark 8.4 should mention this test rather than assert that FK "cover" mixed tori.

Remark 8.7 and Example 8.8. Correct and well hedged. The ledger arithmetic is right: $b = 3$, $\Lambda = 1$ forces $\lambda_\infty = 1/3$, and $(1/3)/9 = 1/27$ balances 27.

Ledger. $N_2 = N_3 = 1$ and $N_5 = 4$ are right.

2. A positivity lemma (recommended for the paper)

Lemma ([**P**]; strengthens Proposition 6.1)**.** *Let $(a_{\rho\tau})$ be the matrix of L_D in the orthonormal character bases (Part III, Theorem 4.3 and Corollary 5.1). The following are equivalent:*

(a) L_D permutes the irreducible characters;

(b) every $a_{\rho\tau}$ is an integer;

(c) every $a_{\rho\tau}$ is real and ≥ 0;

(d) every $L_D\Theta_\rho$ is a positive-definite function on $\widetilde{H}$.

Moreover, for a single ρ, condition (c) for that row (equivalently (d) for that ρ) already implies that $L_D\Theta_\rho$ is a nonnegative combination of characters with norm 1.

Proof. (c)⇒(a). Let $\rho \neq \rho'$. Orthogonality of rows gives $\sum_\tau a_{\rho\tau}a_{\rho'\tau} = 0$, a sum of nonnegative terms. So distinct rows have disjoint supports. Since L_D is unitary, each column has norm 1. A column meets the support of only one row, so its unique nonzero entry is 1. Each row has norm 1 and its nonzero entries are 1, so it has exactly one of them.

(a)⇒(b)⇒(c). (a)⇒(b) is trivial. (b)⇒(c) is Proposition 6.1.

(c)⇔(d). On a group compact modulo the center, with a fixed unitary central character, a continuous class function is positive definite if and only if its coefficients in the irreducible characters are ≥ 0 (by Peter–Weyl and GNS). □

Consequences.

(i) *Integrality is a positivity problem.* Any construction exhibiting $L_D\Theta_\rho$ as the character of an actual, possibly reducible, representation proves irreducibility.

(ii) *A cheaper finite test.* At a finite level, convolution by $L_D\Theta_\rho$ acts on the τ-isotypic part of the twisted group algebra by the scalar $a_{\rho\tau}/\dim\tau$, up to the normalization of measures. So it suffices to check that the Hermitian matrix $\left(L_D\Theta_\rho(h^{-1}g)\right)_{g,h}$ is positive semidefinite. No character table of the cover is needed.

(iii) *A global argument needs only positivity.* In the global argument of §3.5, the exact value of Λ becomes irrelevant; only its positivity matters.

3. Suggestions for the wild integrality problem in general

These are research suggestions, ordered roughly by how directly they address the problem. I mark what I have checked and what is heuristic.

3.1 A clean formula for R_E and a counting identity (checked)

Write $u_{ij} = \lambda_i/\lambda_j$, where the λ_i are the eigenvalues of t. For a field torus $|u_{ij}| = 1$, so $\Delta(t) = \prod_{i<j}|1-u_{ij}|$, with no square roots. Hence

$$R_E(t) = \prod_{i<j}\frac{|1-u_{ij}|}{|1-u_{ij}^n|} = \prod_{i<j}\prod_{\zeta\in\mu_n\smallsetminus 1}|1-\zeta u_{ij}|^{-1},$$

using $1-X^n = \prod_\zeta(1-\zeta X)$, which holds because $\mu_n \subset F$.

The case $r = 2$. Put $u = t/\sigma(t)$. By Hilbert 90 this identifies A_E with E^1, and

$$R_E(u) = \prod_{\zeta \neq 1} |1 - \zeta u|^{-1}.$$

If ζ has order prime to p, then $|1 - \zeta u| = 1$. Indeed, $u \equiv \zeta^{-1}$ would give $N(u) \equiv \zeta^{-2}$, so $\zeta^2 = 1$. Hence only the p-power roots of unity in μ_n contribute.

So for $r = 2$, Step T reduces to comparing $|1-u|$ with $|1-\zeta|$ for $\zeta \in \mu_{p^a}$, along the unit filtration of E^1. The standard behavior of $x \mapsto x^p$ on U^i changes at $i = e/(p-1)$. This should allow Step T to be proved for *all* wild F and all n when $r = 2$, with R_E written explicitly as a combination of indicators of the subgroups $E^1 \cap U_E^k$. I recommend doing this next, since it turns Proposition 6.5 from an example into a theorem.

A counting identity (any r). Let $\Lambda \subset \mathfrak{d}/\mathfrak{e}$ be an $\mathrm{Ad}(t)$-stable lattice, and $N \gg 0$. Then

$$\Delta(t)^{-2} = \#\mathrm{Fix}(\mathrm{Ad}\, t \text{ on } \Lambda/\varpi^N\Lambda), \qquad R_E(t)^2 = \frac{\#\mathrm{Fix}(\mathrm{Ad}\, t^n)}{\#\mathrm{Fix}(\mathrm{Ad}\, t)}.$$

Proof: $|\det(1 - \mathrm{Ad}\, t)|^{-1} = [\Lambda : (1 - \mathrm{Ad}\, t)\Lambda]$, and kernel and cokernel have equal order on the finite module. So R_E^2 is a ratio of permutation-character values. This is the natural form in which to compare R_E with the Frobenius formula of §3.3.

3.2 Gluing via Brauer's characterization (heuristic, but standard in shape)

At a finite level, the question is whether a class function on a finite twisted group is a virtual character. Brauer's characterization theorem says this is detected by restrictions to *elementary* subgroups: cyclic p'-groups times p-groups. By Proposition 6.1, once $L_D\Theta_\rho$ is a virtual character it is irreducible, because it has norm 1 and is positive at 1.

So Step G reduces to elementary subgroups. The p'-parts are handled by tame (Adams-type) arguments. The core is the restriction to the covered pro-p part, $\widetilde{U_D^1}$ together with the p-part of the covered torus. The twisted version of Brauer's theorem should be located in the literature (e.g. Karpilovsky's books on projective representations) or proved. I have not checked that it exists in the form needed.

3.3 Explicit construction: tame types for $D^\times$ and their covers

Since $(n, r) = 1$, a wild place for n has $p \neq r$. So D is *tamely* ramified, and every irreducible representation of $D^\times$ comes from Howe's admissible pairs (E, ξ), with E/F of degree 1 or r. Explicit character formulas are given in Corwin–Howe (Pacific J. Math. 73, 1977), and degrees in Corwin–Moy–Sally.

Proposal.

(1) *Define genuine admissible pairs* $(E, \tilde{\xi})$, with $\tilde{\xi}(zt^*) = \tilde{\theta}(z)\xi(t)$ on the center C_E of $\widetilde{E^\times}$.

(2) *Build* $\tau(E, \tilde{\xi})$ as an induced representation $\mathrm{Ind}_{\tilde{J}}^{\widetilde{H}} \tilde{\Lambda}$, as in Howe's construction. Use the Stone–von Neumann representation of $\widetilde{E^\times}$ on the torus part, and the compact splitting κ^* on deep congruence subgroups, over which the cover splits.

(3) *Compute Θ_τ on elliptic elements by the Frobenius formula* and compare with $L_D\Theta_{\rho(E,\xi)}$. The fixed-point form of R_E^2 in §3.1 is what such a comparison will produce.

This is the route of Bushnell–Henniart's explicit essentially tame Jacquet–Langlands correspondence, and of Zou's simple types for covers (which assume $p \nmid n$). The division-algebra case is the simplest one: no Levi subgroups, and only simple strata. By §2 it suffices to show that each $L_D\Theta_\rho$ is the character of *some* representation.

3.4 Kirillov theory for the deep part (heuristic, with a dimension check)

On a deep congruence subgroup the cover splits. There, Howe's Kirillov theory for compact p-adic groups (Pacific J. Math. 73, 1977), or the orbit method for saturable pro-p groups (González-Sánchez), writes characters as Fourier transforms of coadjoint orbits. In exponential coordinates the nth-power map is the dilation $X \mapsto nX$. So L_D should act by *dilating orbits*. Dilation sends orbits to orbits, which would give irreducible characters directly, and R_E would be the Jacobian correction.

Dimension check. A generic orbit of $D^\times$ has dimension $r(r-1)$. Dilating by n^{-1} multiplies its volume by $|n|^{-r(r-1)}$, so it multiplies the degree by $|n|^{-r(r-1)/2}$. The Heisenberg factor on the torus contributes b_D. Together:

$$b_D\,|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2} = c_D,$$

exactly the predicted degree ratio. This makes the orbit method a promising mechanism for the pro-p part in Step G. The applicability conditions (saturability, p relative to the dimension) need checking.

3.5 The global route (conditional; needs only positivity)

(1) Choose the global D with S containing *every* place above n, together with the target wild place w. Then every finite split place is tame.

(2) Take unit elements at the split finite places. This needs [FK, §12] under $(n, N_r) = 1$, which is automatic for $r \le 5$ with n odd, plus transport to the cover of Part I.

(3) At the tame places of S, Part III, Theorem 7.6, gives permutation matrices.

(4) Globalize ρ (as in [FK, §20]). Take $\tilde{f}_w = e_\tau$ (a projector), so that $f_w = T^{-1}e_\tau$ has $\operatorname{tr}\rho(f_w) = a_{\rho\tau}$.

(5) Separate by the Hecke algebra at split places, and use strong multiplicity one for $D^\times$ (Badulescu). This should give

$$a_{\rho\tau} = (\text{positive constant}) \times (\text{nonnegative integer}).$$

By §2, this already proves integrality, *whatever the constant is*. The remaining inputs are those of Proposition 8.5:

- existence of the split-place and complex-place matching, with positive constants;
- the germ issue at regular x with x^n singular.

Note also: with $S \supseteq \{v \mid n\}$, the product formula gives $\prod_{v\in S} b_{D,v} = \prod_{v|\infty} |n|_v^{(r-1)/2}$. So $\Lambda = 1$ is equivalent to $\lambda_v = |n|_v^{-(r-1)/2}$ at each complex place (compare Example 8.8). But by §2 this exact value is not needed for integrality.

3.6 Order of work

(a) Add the positivity lemma (§2), and fix §1.1 and §1.2.

(b) Run the finite PSD test of §2(ii) for $(2, 3, \mathbb{Q}_3(\zeta_3))$, $\rho = \mathbf{1}$. This is cheaper than a character table.

(c) Prove Step T for all $r = 2$, using the formula of §3.1.

(d) Attack Step G for $r = 2$ by §3.3, the explicit types. This case is small: E is quadratic, and the strata are short.

4. Editorial

- Lemma 6.4 needs a status label.
- The abstract is now long, about 280 words; trim it.
- The acknowledgments sentence "added in Draft 5 after a separate comparison ..." is drafting history. Move it to the log.
- Restore the $\{\pm 1\} \cap \mu_n$ argument alongside "by convention" (Summary).